

Exploratory Data Analysis (EDA) Midterm Proposal

Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning
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This midterm proposal was prepared for ITAI 1371 Machine Learning by group 6171 16347 3. Our team has reviewed and approved this proposal for Prof. Rao's consideration.

GOAL 1: Prepare the Netflix Movies and TV Shows dataset for a future machine learning model that classifies whether a movie may be appropriate for a children's profile based on available movie characteristics.

GOAL 2: Focus the project on movies by filtering out television shows through reproducible Python code.

GOAL 3: Complete the required exploratory data analysis and preprocessing in a reproducible Jupyter/Colab notebook without manually changing the original dataset.

GOAL 4: Demonstrate our understanding of the course material by completing the following requirements:

- 1 Load and inspect the unchanged raw dataset in Python using pandas.
- 2 Create a 70% training and 30% testing split in Python and perform EDA only on the training set.
- 3 Identify and handle NaN and null values using appropriate, documented methods.
- 4 Scale numerical features and normalize skewed distributions when justified by the data.
- 5 Encode useful categorical features, including one-hot encoding where appropriate.
- 6 Examine target-class balance and apply a balancing method only if meaningful imbalance is found.
- 7 Engineer features that support the future classification problem.
- 8 Use Matplotlib to show meaningful before-and-after distributions and print sample records during processing.
- 9 Document the complete pipeline so the cleaned data can be reused for the final ML project.

PROPOSAL

This project will investigate whether machine learning can classify a Netflix movie as potentially appropriate for a children's profile using characteristics such as genre, release year, duration, country, director, cast, and description. The project will prepare and analyze the data through a reproducible Python notebook. The cleaned dataset will later support the development and evaluation of classification models for the final project.

METHODOLOGY

Use the **Netflix Movies and TV Shows** dataset on Kaggle at <https://www.kaggle.com/datasets/shivamb/netflix-shows>. The raw dataset contains 8,807 rows and 12 columns, exceeding the assignment's 2,000-row minimum. Each row represents a Netflix title, with fields including type, title, director, cast, country, date added, release year, rating, duration, listed genres, and description.

- 1 Load the unchanged Kaggle file into Python. Inspect its dimensions, column types, duplicates, missing values, and rating labels without manually editing the source file.
- 2 Use Python to filter the intended project population to movies. After filtering, remove the constant type field because every retained record will have the same value.
- 3 Create the binary kid_safe target from the rating field. Code G and PG as 1 and clearly mature ratings as 0. Missing or unrated observations will be separated for human review rather than automatically labeled unsafe.
- 4 Remove the original rating field from the model inputs after constructing the target. Using rating as both the source of the target and a predictor would create target leakage.
- 5 Split the prepared movie population programmatically into 70% training and 30% testing sets. Perform EDA and determine preprocessing choices only with the training data. Keep the test data untouched for future evaluation.
- 6 Identify and handle NaN and null values using methods appropriate to each feature. Document whether values are filled, assigned an explicit unknown category, or excluded and explain why.
- 7 Encode useful categorical input features such as country and genre, including one-hot encoding where appropriate. Scale numerical features and normalize skewed distributions only when those transformations are supported by the training-data analysis.
- 8 Engineer potentially useful features from duration, release year, country, genre, director, cast, and description. Exclude identifiers and constant or leakage-producing fields from the future model inputs.
- 9 Use Matplotlib to compare meaningful distributions before and after preprocessing. Print sample rows after major stages so the notebook clearly demonstrates how the data changes.
- 10 Examine the kid_safe target distribution in the training set. If meaningful imbalance is identified, apply an appropriate balancing technique only to the training data, not to the untouched test data.
- 11 Export the final cleaned dataset and upload the required materials to GitHub by 03-Oct-2026: the original dataset URL document, proposal PDF, Jupyter notebook, cleaned dataset, and a separate reflection/contribution journal containing an entry from every team member.

EXPECTED RESULTS

We expect the movie-only dataset to remain above the assignment's 2,000-row minimum after television shows are removed through Python. The training-data analysis will likely identify missing values in fields such as director, cast, country, date added, rating, and duration. Missing and unrated observations will be described accurately rather than treated as statistical outliers.

We also expect the engineered `kid_safe` target to be imbalanced because G- and PG-rated movies may be less common than mature-rated movies. The notebook will measure the actual class distribution before selecting a response. If balancing is needed, it will be applied only to the training data so the test set continues to represent unseen data.

The completed notebook will document every preprocessing decision and display sample records before and after major processing stages. Matplotlib visualizations will compare meaningful distributions, such as missing-value patterns, numerical-feature distributions, and the target distribution before and after any justified class-balancing procedure. All transformations will be completed through Python without manually changing the original Kaggle file.

The final output will be a cleaned, reproducible dataset with appropriate missing-value handling, encoded categorical variables, transformed numerical features where necessary, and engineered features derived from the available movie information. The original rating column will not be included among the model inputs because it is used to construct the prediction target.

This prepared dataset will support the group's final binary-classification project. The untouched 30% test set will be retained for future model evaluation, and each group member's work will be documented separately in the required reflection and contribution journal.